김인녕

A. 교수님과 연구실 소개

교수님 약력

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- POSTECH 기계공학과 석·박사통합 (20′~25′)
- POSTECH 기계공학과 학사 (15′~20′)
- 대한기계학회 기계인공지능연구회 전문위원

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  (**I**ntelligent **F**low **S**imulation & **O**ptimization Lab)
- Google Scholar:
  https://scholar.google.com/citations?user=nL8zSVkAAAAJ&hl=ko
- 연구실 홈페이지: *https://sites.google.com/view/innyounglab*



대표 연구 키워드

Multiphysics Simulation

- 난류, 다상, 고체-유체 연성,
소음 등 복합물리 정밀 해석

Autonomous CFD

- 강화학습 기반 격자 생성 및
수렴 유동 해석 자동화

Optimal Design & Control

- 강화학습 기반 다변수, 고비용,
비선형 시스템 최적 설계 및 제어

Agentic CAE

- Agentic AI 기반 공학 해석·설계
과정의 통합 자율화

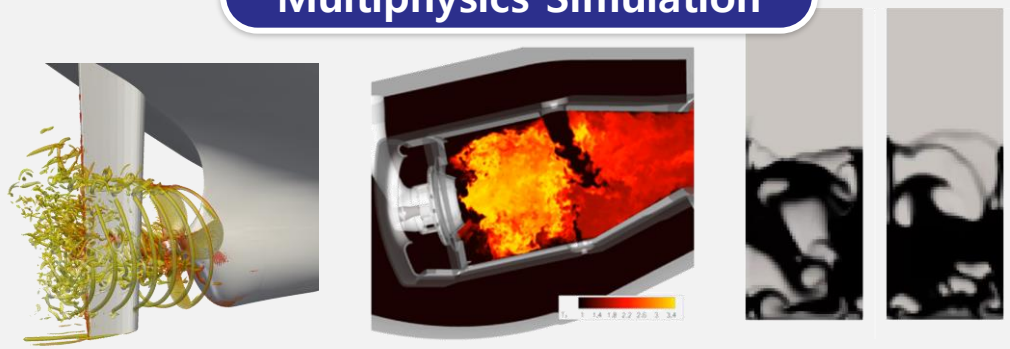
기계 인공 지능 연구회

산학 협력 탐색 그룹

김인녕

B. 주요 연구성과 (논문 중심)

Multiphysics Simulation

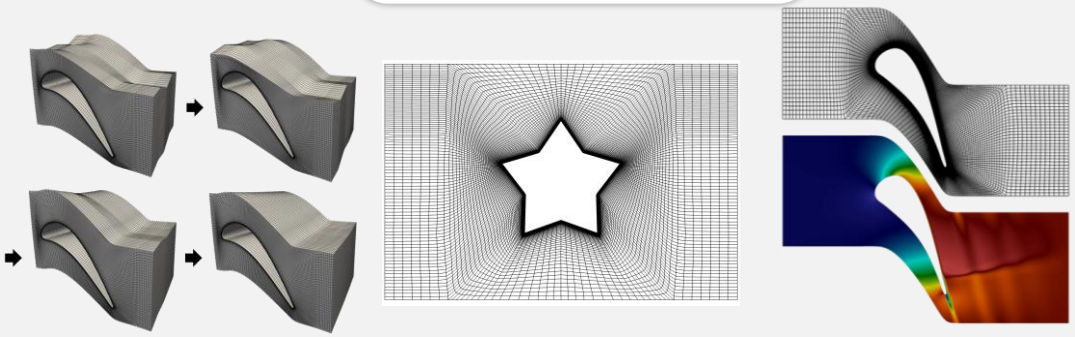


Turbulence, Heat Transfer, Multiphase, …

- **Kim, I.**, et al. (2023). Domain reduction strategy for large-eddy simulation to predict underwater radiated noise from a marine propeller. *Ocean Eng.*, *279*, 114538. · **JCR Top 5.4%**
- Jeon, Y.*, **Kim, I.***, et al. (2024). Numerical analysis of flow-induced noise during the charging of a fuel cell electric vehicle. *J. Mech. Sci. Technol.*, *38*(11), 6129–6139.
- Kwon, J., **Kim, I.**, et al. (2026). Tracing and healing multiscale defects for cost-effective lithium carbonate-derived layered oxide cathodes. *Adv. Energy Mater.*, e71727. · **JCR Top 2.5%**

*Co-first authors

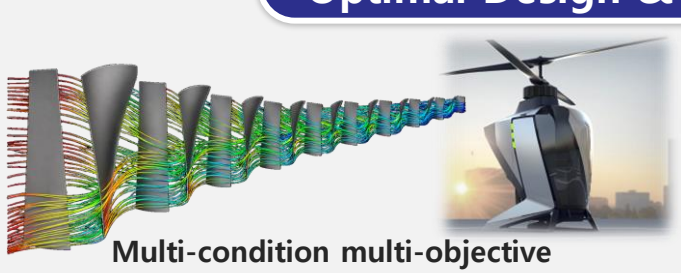
Autonomous CFD



Meshing Simulation

- **Kim, I.**, et al. (2024). Non-iterative generation of an optimal mesh for a blade passage using deep reinforcement learning. *Comput. Phys. Commun.*, *294*, 108962. · **JCR Top 0.8%**
- **Kim, I.**, et al. (2025). Automatic mesh generation for optimal CFD of a blade passage using deep reinforcement learning. *J. Comput. Phys.*, *541*, 114306. · **JCR Top 2.5%**

Optimal Design & Control



Multi-condition multi-objective high-dimensional design

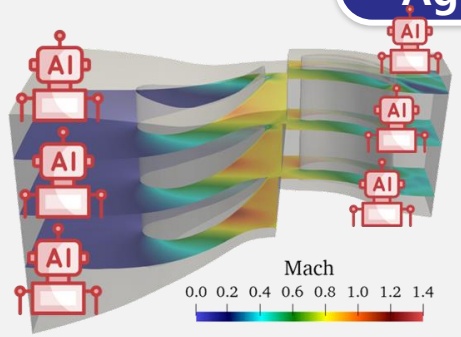


Bio-inspired flyer control

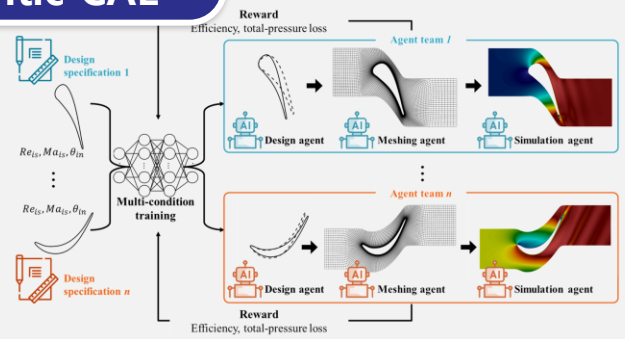
- Kim, S.*, **Kim, I.***, & You, D. (2022). Multi-condition multi-objective optimization using deep reinforcement learning. *J. Comput. Phys.*, *462*, 111263. · **JCR Top 4.5%**
- **Kim, I.**, & You, D. (2025). Discovering optimal gas injection strategies for a fluidized bed system using deep reinforcement learning. *Phys. Fluids*, *37*(8). · **JCR Top 6.1%**
- Hong, S.*, Kim, S.*, **Kim, I.***, & You, D. (2025). Deep reinforcement learning-based control algorithm for flight kinematics of insect-scale flyers. *Adv. Eng. Softw.*, *210*, 104014. · **JCR Top 8.9%**

*Co-first authors

Agentic CAE



Cooperative agents for system-level design



Agent teams for end-to-end simulation & design

- **Kim, I.**, & You, D. (2026). Automation of the CFD-based design process for turbomachinery blades using deep reinforcement learning. *Comput. Fluids*, *317*, 107163.